PROJECT

Stock Price Prediction

A Comprehensive Technical Report

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Declaration

I hereby declare that the project titled **"Stock Price Prediction"** submitted by me is a genuine and original piece of academic work carried out under the guidance and supervision of my faculty mentor. This project report has been prepared by me as part of the academic requirements for my course, and it reflects my own understanding, effort, and implementation of the subject matter.

I further declare that all the work presented in this report—including problem formulation, data preprocessing, feature engineering, model building using PySpark, analysis, visualization, and interpretation of results—has been carried out by me. Wherever the work of others has been used or referred to, it has been properly acknowledged and cited in accordance with academic standards.

This project has not been submitted, either wholly or partially, for the award of any other degree, diploma, or certification at this or any other institution. I take full responsibility for the authenticity and accuracy of the content presented in this report.

I also confirm that all datasets, tools, libraries, and external resources used in this project have been appropriately credited in the Bibliography section. Any resemblance to existing works, if found, is purely coincidental and unintentional.

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Abstract

Stock price prediction remains one of the most challenging and significant applications of data analytics in the financial domain. The stock market generates large volumes of structured time-dependent data in the form of daily opening prices, highest prices, lowest prices, closing prices, adjusted closing prices, and trading volume. Analysing such historical market data manually is insufficient for identifying hidden patterns, especially when the size and frequency of data continue to increase. This has created strong demand for scalable computational systems capable of performing data engineering, feature extraction, model training, and visual analysis within a unified pipeline.

This report presents the design, implementation, and evaluation of a **Stock Price Prediction System** developed using **Apache Spark (PySpark 3.x)** and the **MLlib Linear Regression algorithm**. The system uses real historical stock market data obtained from a Kaggle dataset, specifically a CSV file for Apple Inc. stock data, containing attributes such as Date, Open, High, Low, Close, Adj Close, and Volume. A structured preprocessing pipeline is applied to the dataset, including schema inference, date conversion, missing value checking, mean-based imputation for numeric fields, ordering of records by date, and creation of derived time-aware features.

To improve predictive power, the project performs feature engineering using **lag-based features** such as previous day closing price, two-day lag closing price, and previous day volume, along with a **3-day moving average** of the closing price. These engineered features are combined into a machine-learning-ready vector using **VectorAssembler**. The dataset is then split into training and testing subsets in an 80/20 ratio, and a Linear Regression model is trained to predict the stock closing price.

Model evaluation is carried out using **RMSE (Root Mean Squared Error)** and **R^2 Score**, which quantify the prediction error and model fit respectively. In addition to numerical metrics, the system produces three attractive visualisations: an **Actual Stock Closing Price Trend**, a **Predicted Stock Closing Price Trend**, and an **Actual vs Predicted Overlay Graph**. These graphs improve interpretability and help demonstrate how closely the model follows real market behaviour.

This project highlights the value of PySpark in building scalable and reproducible machine learning workflows for financial forecasting. It also serves as a foundation for future extensions such as advanced regression models, time-series learning methods, multi-stock analysis, cloud deployment, and real-time financial dashboards.

Keywords: Apache Spark, PySpark, MLlib, Linear Regression, Stock Price Prediction, Feature Engineering, Lag Features, Moving Average, RMSE, R^2 Score, Big Data Analytics, Financial Forecasting, Data Engineering

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Chapter 1: Introduction

1.1 Background and Motivation

The stock market is one of the most dynamic and influential systems in the modern global economy. Every trading session generates a continuous stream of structured financial data containing stock prices, transaction volume, and time-based market changes. Investors and analysts attempt to use this information to forecast future movements and support decision-making. However, manual observation of charts and spreadsheets becomes increasingly inefficient when datasets grow in size and complexity.

The intersection of big data technologies and machine learning offers a practical solution to this problem. With the help of distributed data processing frameworks, it becomes possible to build systems that are not only accurate for small academic datasets but are also designed to scale for large production-level workloads. This project operates within that intersection by applying **PySpark** to historical stock market data and constructing a structured machine learning workflow for closing-price prediction.

The motivation behind this work is both academic and practical. Academically, it demonstrates core data engineering skills such as schema inference, preprocessing, feature engineering, and model evaluation using PySpark. Practically, it shows how historical market data can be transformed into useful predictive insights using computational techniques.

1.2 Importance of Stock Price Prediction

Stock price prediction is important because it supports investment planning, trend analysis, portfolio management, and risk reduction. Even if the stock market is not perfectly predictable, identifying patterns in historical data can still provide value for short-term estimation and analytical understanding.

A stock's closing price is especially significant because it is widely used in technical analysis, market summaries, and decision-making processes. Predicting closing prices using machine learning helps demonstrate how historical price behaviour and volume information can be converted into measurable forecasting outputs.

1.3 The Role of Big Data in Financial Analytics

Modern financial analytics increasingly depends on computational systems that can process large datasets quickly and efficiently. Traditional spreadsheet tools and single-machine scripts may work for small datasets, but they become slower and less scalable as the amount of data increases.

Apache Spark addresses these limitations through distributed execution, in-memory computation, and optimised query planning. Spark's DataFrame API and MLlib library make it possible to combine data engineering and machine learning in the same environment. This is particularly useful in financial analytics, where data cleaning, transformation, modelling, and interpretation are closely connected.

1.4 Project Objectives

The primary objectives of this project are as follows:

- To load real historical stock market data into a Spark DataFrame using PySpark
- To clean and preprocess the dataset by checking and handling missing values
- To engineer time-based features such as lag values and moving averages
- To assemble all predictor variables into a feature vector using VectorAssembler
- To train a Linear Regression model for predicting stock closing prices
- To evaluate the model using RMSE and R^2 score
- To generate multiple visualisations for actual and predicted stock trends
- To prepare a formal technical report documenting the complete pipeline

1.5 Scope and Limitations

This project is developed as an academic proof-of-concept focused on demonstrating the use of PySpark for financial prediction. The system is designed around a single real stock dataset and uses a Linear Regression model to forecast closing prices.

The main limitations are:

- The model does not incorporate external market signals such as news, company announcements, or investor sentiment
- Only one stock symbol is used for simplicity
- Linear Regression is a baseline algorithm and may not capture highly nonlinear market behaviour
- The project does not include real-time streaming or cloud deployment

These limitations do not reduce its value as a practical big-data machine-learning pipeline, but they define the boundaries of its current scope.

Chapter 2: Literature Review and Problem Statement

2.1 Traditional Stock Market Analysis Approaches

Traditional stock analysis is generally divided into **technical analysis** and **fundamental analysis**. Technical analysis studies price patterns, support and resistance levels, and trend indicators, while fundamental analysis examines financial statements, management quality, revenue, and macroeconomic conditions. Both methods are useful, but they often rely heavily on manual reasoning and experience.

In academic and industrial settings, there has been increasing interest in replacing or supporting these traditional approaches with data-driven predictive models that can process large historical datasets.

2.2 Machine Learning in Financial Forecasting

Machine learning has become a significant area of research in stock market prediction because of its ability to identify relationships within complex numerical data. Regression models are commonly used when the objective is to estimate a continuous value such as stock price. Other methods such as Random Forest, Support Vector Machines, Gradient Boosting, and Neural Networks have also been explored for stock forecasting.

Among these, Linear Regression remains an important baseline model because it is simple, interpretable, and easy to implement. It helps establish a reference point before moving to more advanced algorithms.

2.3 Limitations of Conventional Tools

Traditional tools face several challenges when applied to financial datasets:

- They are often limited by system memory
- They are not designed for distributed execution
- They become slow with repeated transformations and larger data volumes
- They make it difficult to scale from academic experiments to enterprise-grade systems

These issues motivate the use of frameworks such as Spark that are designed for scalable analytics.

2.4 Why Big Data Frameworks are Needed

Big data frameworks are needed because financial systems produce high-volume and high-frequency data. Apache Spark offers a practical platform for this by combining:

- distributed data handling
- schema-aware transformations
- machine learning support
- fault-tolerant execution

This enables the same program design to handle both small project datasets and much larger industrial datasets with minimal structural changes.

2.5 Problem Statement

The problem addressed in this project can be stated as follows:

How can historical stock market data be processed using PySpark in a scalable, reproducible, and efficient manner to predict stock closing prices through Linear Regression, and how can the resulting predictions be analysed using meaningful visualisations?

Chapter 3: Methodology

3.1 Dataset Description

The dataset used in this project is a real historical stock dataset obtained from Kaggle. The selected file contains Apple Inc. stock data and includes the following columns:

- Date
- Open
- High
- Low
- Close
- Adj Close
- Volume

For this project, only the following columns are used in the predictive pipeline:

- Date
- Open
- High
- Low
- Close
- Volume

The Adj Close column is excluded because the project is specifically designed around direct closing-price prediction using a defined set of core market variables.

3.2 Apache Spark Architecture and Data Ingestion

Apache Spark is a fast distributed data processing engine designed for scalable analytics. In this project, Spark is accessed through PySpark and a local SparkSession is created to initialise the environment.

The dataset is ingested using:

```
df = spark.read.csv("stock_data.csv", header=True, inferSchema=True)
```

The header=True parameter ensures that column names are read from the first row, while inferSchema=True allows Spark to infer the appropriate data types automatically.

3.3 Data Cleaning and Missing Value Strategy

After ingestion, the first preprocessing step is to inspect the data for missing values. Missing values in numeric columns can disrupt machine learning models, so the project checks each relevant field and fills missing entries using the mean of that column. Mean imputation is simple, reproducible, and suitable for a baseline project when missingness is not extensive.

The Date column is converted into a proper Spark date type using to_date(), and invalid date rows are removed.

3.4 Feature Engineering with Lag and Moving Average

Feature engineering is central to this project because raw OHLCV data alone may not fully represent short-term price dependencies. The following derived features are created:

- Close_lag_1: closing price of the previous day
- Close_lag_2: closing price two days earlier
- Volume_lag_1: previous day trading volume
- Close_MA_3: 3-day moving average of closing price

These features allow the model to include short-range time dependence and recent trend information.

3.5 Feature Vector Construction using VectorAssembler

PySpark MLlib requires all predictor variables to be assembled into a single vector column. This is achieved using VectorAssembler, which combines the following fields:

- Open
- High
- Low
- Volume
- Close_lag_1
- Close_lag_2
- Volume_lag_1
- Close_MA_3

The assembled vector becomes the input feature set for the Linear Regression model.

3.6 Linear Regression Theory and Parameters

Linear Regression is a supervised learning algorithm used for predicting continuous outcomes. It estimates the relationship between a dependent variable and one or more independent variables. In this project:

- **Label/Target** = Close
- **Features** = assembled numerical feature vector

The final dataset is split into:

- 80% training data
- 20% testing data

A reproducible random seed is used during splitting to maintain consistency in repeated runs.

Chapter 4: Implementation

4.1 System Architecture Overview

The project is implemented as a sequential PySpark workflow that includes:

1. Data ingestion
2. Preprocessing
3. Feature engineering
4. Feature vector creation
5. Model training
6. Prediction
7. Evaluation
8. Visualisation

This architecture is easy to explain, easy to reproduce, and well-suited to an academic practical project.

4.2 Data Processing Pipeline

The script begins by creating a Spark session and loading the real stock data CSV file. It then filters required columns, converts the Date column, checks missing values, fills missing numeric values, and sorts the dataset by date. Spark window functions are used to compute lag-based features and a moving average feature.

4.3 Model Training and Prediction Workflow

After preprocessing, the input columns are combined into a feature vector with VectorAssembler. The resulting DataFrame is then split into training and test sets, and a LinearRegression model is trained on the training subset. Predictions are generated on the testing subset and sorted by date to preserve trend interpretation.

4.4 Visualisation Design

The project creates three separate visual outputs:

1. **Actual Stock Closing Price Trend**
2. **Predicted Stock Closing Price Trend**
3. **Actual vs Predicted Overlay Graph**

These visualisations are designed to improve clarity. The separate graphs isolate actual and predicted patterns individually, while the overlay graph combines both into a single comparative chart.

Chapter 5: Results and Discussion

5.1 Model Performance Metrics

The model is evaluated using two important regression metrics:

- **RMSE (Root Mean Squared Error)**
- **R^2 Score**

```
==================================================
                MODEL EVALUATION
==================================================
RMSE     : 1.8105112349757914e-10
R2 Score : 1.0
```

RMSE measures the average magnitude of prediction error, while R^2 indicates how well the model explains variation in the target variable.

5.2 Interpretation of RMSE and R^2

A lower RMSE indicates that predictions are closer to the real values. An R^2 score closer to 1 indicates that the model has a stronger fit to the observed data. Together, these two metrics provide a balanced view of model quality.

5.3 Analysis of Actual Stock Price Trend

The actual trend graph displays the real closing prices from the test dataset. This graph helps visualise the genuine market movement over time and serves as the baseline against which predictions are compared.

5.4 Analysis of Predicted Stock Price Trend

The predicted trend graph displays the output generated by the Linear Regression model. It helps reveal whether the model captures the general direction and variation of stock movement during the selected testing period.

5.5 Comparative Analysis using Overlay Graph

The overlay graph is the most informative of the three visualisations because it shows both actual and predicted prices together. If the two lines remain close across most dates, it suggests that the model is able to follow the market pattern effectively. Any visible divergence indicates areas where the prediction error is larger.

This visual comparison complements RMSE and R^2 by making model behaviour easier to understand.

Chapter 6: Conclusion and Future Work

6.1 Summary of Achievements

This project successfully demonstrates the application of PySpark to stock price prediction using real historical market data. The complete pipeline includes data loading, preprocessing, feature engineering, vector assembly, model training, prediction, evaluation, and visualisation. The use of Linear Regression makes the project simple, interpretable, and suitable for academic demonstration.

The project also shows how Spark can serve as a scalable foundation for financial machine learning pipelines.

6.2 Limitations

The project has the following limitations:

- It uses only one stock dataset
- It does not incorporate external financial or macroeconomic variables
- It uses a baseline Linear Regression model rather than more sophisticated forecasting algorithms
- It does not include live market data or streaming support

6.3 Future Scope

Future work can extend this system in several directions:

- Use multiple stock symbols to build a broader forecasting system
- Add technical indicators such as RSI, MACD, and Bollinger Bands
- Apply advanced models such as Random Forest Regressor, XGBoost, or LSTM
- Implement time-series-aware train/test splitting
- Build a dashboard for interactive analytics
- Deploy the pipeline on cloud infrastructure for larger-scale processing

GitHub Repository

https://github.com/pratikyadav0/Stock-Price-Prediction-Using-PySpark.git

The repository contains the following files and directories:

Stock-Price-Prediction

stock_price_prediction.py

requirements.txt

stock_data.csv

actual_stock_price.png

predicted_stock_price.png

actual_vs_predicted_overlay.png

Measures.png

Stock_Price_Prediction_Presentation.pptx

Stock_Price_Prediction_Report.pdf

README.md

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